

Học phần: ĐỌC – VIẾT B2.2 (Reading & Writing B2.2)  
Thời gian làm bài: 40 phút

MÃ ĐỀ  
**04**

Không sử dụng tài liệu [x] Được sử dụng tài liệu [ ] Nộp lại đề thi [x]

Đề thi gồm 25 câu, 2 bài đọc (Passage 1, Passage 2), kèm 01 Phiếu trả lời ở trang cuối. Cán bộ coi thi không giải thích gì thêm.

Sinh viên ghi **MÃ ĐỀ** và làm bài trên **PHIẾU TRẢ LỜI**.

**PASSAGE 1 (Questions 1–12)** 12 questions · about 20 minutes

**Leaf-Cutting Ants and Fungus**

**A** The ants and their agriculture have been extensively studied over the years, but the recent research has uncovered intriguing new findings about the fungus they cultivate, how they domesticated it and how they cultivate it and preserve it from pathogens. For example, the fungus farms, which the ants were thought to keep free of pathogens, turn out to be vulnerable to a devastating mold, found nowhere else but in ants' nests. To keep the mold in check, the ants long ago made a discovery that would do credit to any pharmaceutical laboratory.

**B** Leaf-cutting ants and their fungus farms are a marvel of nature and perhaps the best known example of symbiosis, the mutual dependence of two species. The ant's achievement is remarkable – the biologist Edward O. Wilson has called it “one of the major breakthroughs in animal evolution” – because it allows them to eat, courtesy of their mushroom's digestive powers, the otherwise poisoned harvest of tropical forests whose leaves are laden with terpenoids, alkaloids and other chemicals designed to sicken browsers.

**C** Fungus growing seems to have originated only once in evolution, because all gardening ants belong to a single tribe, the descendants of the first fungus farmer. There are more than 200 known species of the attine ant tribe, divided into 12 groups, or genera. The leaf-cutters use fresh vegetation; the other groups, known as the lower attines because their nests are smaller and their techniques more primitive, feed their gardens with detritus like dead leaves, insects and feces.

**D** The leaf-cutters' fungus was indeed descended from a single strain, propagated clonally, or just by budding, for at least 23 million years. But the lower attine ants used different varieties of the fungus, and in one case a quite separate species, the four biologists discovered. The pure strain of fungus grown by the leaf-cutters, it seemed to Mr. Currie, resembled the monocultures of various human crops, that are very productive for a while and then succumb to some disastrous pathogen, such as the Irish potato blight. Monocultures, which lack the genetic diversity to respond to changing environmental threats, are sitting ducks for parasites. Mr. Currie felt there had to be a parasite in the ant-fungus system. But a century of ant research offered no support for the idea. Textbooks describe how leaf-cutter ants scrupulously weed their gardens of all foreign organisms. “People kept telling me, ‘You know the ants keep their gardens free of parasites, don't you?’” Mr. Currie said of his efforts to find a hidden interloper.

**E** But after three years of sifting through attine ant gardens, Mr. Currie discovered they are far from free of infection. In last month's issue of the *Proceeding of the National Academy of Sciences*, he and two colleagues, Dr. Mueller and David Mairoch, isolated several alien organisms, particularly a family of parasitic molds called *Escovopsis*.

**F** *Escovopsis* turns out to be a highly virulent pathogen that can devastate a fungus garden in a couple of days. It blooms like a white cloud, with the garden dimly visible underneath. In a day or two the whole garden

is enveloped. "Other ants won't go near it and the ants associated with the garden just starve to death," Dr. Rehner said. "They just seem to give up, except for those that have rescued their larvae." The deadly mold then turns greenish-brown as it enters its spore-forming stage.

**G** Evidently the ants usually manage to keep Escovopsis and other parasites under control. But with any lapse in control, or if the ants are removed, Escovopsis will quickly burst forth. Although new leaf-cutter gardens start off free of Escovopsis, within two years some 60 percent become infected. The discovery of Escovopsis's role brings a new level of understanding to the evolution of the attine ants. "In the last decade, evolutionary biologists have been increasingly aware of the role of parasites as driving forces in evolution," Dr. Schultz said. There is now a possible reason to explain why the lower attine species keep changing the variety of fungus in their mushroom gardens, and occasionally domesticating new ones – to stay one step ahead of the relentless Escovopsis.

**H** Interestingly, Mr. Currie found that the leaf-cutters had in general fewer alien molds in their gardens than the lower attines, yet they had more Escovopsis infections. It seems that the price they pay for cultivating a pure variety of fungus is a higher risk from Escovopsis. But the leaf-cutters may have little alternative: they cultivate a special variety of fungus which, unlike those grown by the lower attines, produces nutritious swollen tips for the ants to eat.

**I** Discovery of a third partner in the ant-fungus symbiosis raises the question of how the attine ants, especially the leaf-cutters, keep this dangerous interloper under control. Amazingly enough, Mr. Currie has again provided the answer. "People have known for a hundred years that ants have a whitish growth on the cuticle," said Dr. Mueller, referring to the insects' body surface. "People would say this is like a cuticular wax. But Cameron was the first one in a hundred years to put these things under a microscope. He found it was not inert wax. It is alive." Mr. Currie discovered a specialized patch on the ants' cuticle that harbors a particular kind of bacterium, one well known to the pharmaceutical industry, because it is the source of half the antibiotics used in medicine. From each of 22 species of attine ant studied, Mr. Cameron and colleagues isolated a species of Streptomyces bacterium, they reported in Nature in April. The Streptomyces does not have much effect on ordinary laboratory funguses. But it is a potent poisoner of Escovopsis, inhibiting its growth and suppressing spore formation. It also stimulates growth of the ants' mushroom fungus. The bacterium is carried by virgin queens when they leave to establish new nests, but is not found on male ants, playboys who take no responsibility in nest-making or gardening.

**J** Because both the leaf-cutters and the lower attines use Streptomyces, the bacterium may have been part of their symbiosis for almost as long as the Escovopsis mold. If so, some Alexander Fleming of an ant discovered antibiotics millions of years before people did. Even now, the ants are accomplishing two feats beyond the powers of human technology. The leaf-cutters are growing a monocultural crop year after year without disaster, and they are using an antibiotic apparently so wisely and prudently that, unlike people, they are not provoking antibiotic resistance in the target pathogen.

**Questions 1–6**

*Use the information in the passage to match the options (A–C) with the activities or features of ants below. You may use any letter more than once.*

|                            |                        |               |
|----------------------------|------------------------|---------------|
| <b>A</b> Leaf-cutting ants | <b>B</b> Lower attines | <b>C</b> Both |
|----------------------------|------------------------|---------------|

- 1. Build small nests and live with different foreign fungus. \_\_\_\_\_
- 2. Use toxic leaves to feed fungus. \_\_\_\_\_
- 3. Raise fungus which don't live with other foreigners. \_\_\_\_\_
- 4. Use substance to fight against Escovopsis. \_\_\_\_\_
- 5. Use dead vegetable to feed fungus. \_\_\_\_\_
- 6. Are free of parasites, as explained previously. \_\_\_\_\_

### Questions 7–11

*The passage has ten paragraphs, A–J. Which paragraph contains the following information? Write the correct letter, A–J.*

7. Dangerous outcome of Escovopsis. \_\_\_\_\_
8. Disadvantage of growing a single fungus. \_\_\_\_\_
9. Comparison of features of two different nests. \_\_\_\_\_
10. Two achievements made by ants earlier than humans. \_\_\_\_\_
11. Advantage of growing a new breed of fungus. \_\_\_\_\_

### Question 12

*Choose the correct letter, A, B, C or D.*

12. How does the author think of Currie's opinion?
  - A. His viewpoint was verified later.
  - B. Earlier study has sufficient evidence.
  - C. No details mentioned in the article.
  - D. His opinion was proved to be wrong.

## Assessing the risk

**A** As a title for a supposedly unprejudiced debate on scientific progress, “Panic attack: interrogating our obsession with risk” did not bode well. Held last week at the Royal Institution in London, the event brought together scientists from across the world to ask why society is so obsessed with risk and to call for a “more rational” approach. “We seem to be organising society around the grandmotherly maxim of ‘better safe than sorry’,” exclaimed Spiked, the online publication that organised the event. “What are the consequences of this overbearing concern with risks?”

**B** The debate was preceded by a survey of 40 scientists who were invited to describe how awful our lives would be if the “precautionary principle” had been allowed to prevail in the past. Their response was: no heart surgery or antibiotics, and hardly any drugs at all; no aeroplanes, bicycles or high-voltage power grids; no pasteurisation, pesticides or biotechnology; no quantum mechanics; no wheel; no “discovery” of America. In short, their message was: no risk, no gain.

**C** They have absolutely missed the point. The precautionary principle is a subtle idea. It has various forms, but all of them generally include some notion of cost-effectiveness. Thus the point is not simply to ban things that are not known to be absolutely safe. Rather, it says: “Of course you can make no progress without risk. But if there is no obvious gain from taking the risk, then don’t take it.”

**D** Clearly, all the technologies listed by the 40 well-chosen savants were innately risky at their inception, as all technologies are. But all of them would have received the green light under the precautionary principle because they all had the potential to offer tremendous benefits – the solutions to very big problems – if only the snags could be overcome.

**E** If the precautionary principle had been in place, the scientists tell us, we would not have antibiotics. But of course we would – if the version of the principle that sensible people now understand had been applied. When penicillin was discovered in the 1920s, infective bacteria were laying waste to the world. Children died from diphtheria and whooping cough, every open drain brought the threat of typhoid, and any wound could lead to septicaemia and even gangrene.

**F** Penicillin was turned into a practical drug during the Second World War, when the many pestilences that result from war threatened to kill more people than the bombs. Of course antibiotics were a priority. Of course the risks, such as they could be perceived, were worth taking.

**G** And so with the other items on the scientists’ list: electric light bulbs, blood transfusions, CAT scans, knives, the measles vaccine – the precautionary principle would have prevented all of them, they tell us. But this is just plain wrong. If the precautionary principle had been applied properly, all these creations would have passed muster, because all offered incomparable advantages compared to the risks perceived at the time.

**H** Another issue is at stake here. Statistics are not the only concept people use when weighing up risk. Human beings, subtle and evolved creatures that we are, do not survive to three-score years and ten simply by thinking like pocket calculators. A crucial issue is consumer’s choice. In deciding whether to pursue the development of a new technology, the consumer’s right to choose should be considered alongside considerations of risk and benefit. Clearly, skiing is more dangerous than genetically modified tomatoes. But people who ski choose to do so; they do not have skiing thrust upon them by portentous experts of the kind who now feel they have the right to reconstruct our crops. Even with skiing there is the matter of cost effectiveness to consider: skiing, I am told, is exhilarating. Where is the exhilaration in GM soya?

**I** Indeed, in contrast to all the other items on Spiked’s list, GM crops stand out as an example of a technology whose benefits are far from clear. Some of the risks can at least be defined. But in the present economic climate, the benefits that might accrue from them seem dubious. Promoters of GM crops believe

that the future population of the world cannot be fed without them. That is untrue. The crops that really matter are wheat and rice, and there is no GM research in the pipeline that will seriously affect the yield of either. GM is used to make production cheaper and hence more profitable, which is an extremely questionable ambition.

**J** The precautionary principle provides the world with a very important safeguard. If it had been in place in the past it might, for example, have prevented insouciant miners from polluting major rivers with mercury. We have come to a sorry pass when scientists, who should above all be dispassionate scholars, feel they should misrepresent such a principle for the purposes of commercial and political propaganda. People at large continue to mistrust science and the high technologies it produces partly because they doubt the wisdom of scientists. On such evidence as this, these doubts are fully justified.

### Questions 13–18

*Do the following statements agree with the information given in the passage? Write TRUE if the statement is true, FALSE if the statement is false, NOT GIVEN if the information is not given.*

13. The title of the debate is not unbiased. \_\_\_\_\_
14. All the scientists invited to the debate were from the field of medicine. \_\_\_\_\_
15. The message the scientists who conducted the survey were sending was that people shouldn't take risks. \_\_\_\_\_
16. All 40 listed technologies are riskier than other technologies. \_\_\_\_\_
17. It was worth taking the risks to invent antibiotics. \_\_\_\_\_
18. All the other inventions on the list were also judged by the precautionary principle. \_\_\_\_\_

### Questions 19–25

*Complete the summary below. Use NO MORE THAN THREE WORDS from the passage for each answer.*

When applying the precautionary principle to decide whether to invent a new technology, people should also take into consideration the (19) \_\_\_\_\_, along with the usual consideration of (20) \_\_\_\_\_. For example, though risky and dangerous enough, people still enjoy (21) \_\_\_\_\_ for the excitement it provides. On the other hand, experts believe that the future population desperately needs (22) \_\_\_\_\_ in spite of their undefined risks. However, the research conducted so far has not been directed towards increasing the yield of (23) \_\_\_\_\_, but to reduce the cost of (24) \_\_\_\_\_ and to bring more profit out of it. In the end, such selfish use of the precautionary principle for business and political gain has often led people to (25) \_\_\_\_\_ science, for they believe scientists are not to be trusted.

— END OF THE TEST —

TP. Hồ Chí Minh, ngày ..... tháng ..... năm 2026

**Trưởng Bộ môn duyệt**  
(Ký và ghi rõ họ, tên)

TP. Hồ Chí Minh, ngày ..... tháng ..... năm 2026

**Giảng viên ra đề**  
(Ký và ghi rõ họ, tên)

HỌC VIỆN HÀNG KHÔNG VIỆT NAM · KHOA NGOẠI NGỮ  
**PHIẾU TRẢ LỜI / STUDENT ANSWER SHEET**

Kiểm tra giữa kỳ · ĐỌC – VIẾT B2.2 (READING) · Năm học 2026 – 2027 · Học kỳ 1

|                                                |                           |
|------------------------------------------------|---------------------------|
| Họ và tên: .....                               | <b>MÃ ĐỀ</b><br><b>04</b> |
| MSSV: ..... Lớp: .....                         |                           |
| Phòng thi: ..... Số tờ: ..... Chữ ký SV: ..... |                           |

| TỔNG (/10) | Số câu đúng (/25) | GV chấm 1 | GV chấm 2 |
|------------|-------------------|-----------|-----------|
|            |                   |           |           |

**ANSWERS** — Write *ONLY* the answer (letter, Roman numeral, word(s) or TRUE / FALSE / NOT GIVEN / YES / NO).  
Spelling must be correct. 0.4 point per question.

| Câu | Your answer | Câu | Your answer |
|-----|-------------|-----|-------------|
| 1.  |             | 14. |             |
| 2.  |             | 15. |             |
| 3.  |             | 16. |             |
| 4.  |             | 17. |             |
| 5.  |             | 18. |             |
| 6.  |             | 19. |             |
| 7.  |             | 20. |             |
| 8.  |             | 21. |             |
| 9.  |             | 22. |             |
| 10. |             | 23. |             |
| 11. |             | 24. |             |
| 12. |             | 25. |             |
| 13. |             |     |             |